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A novel framework with ComMAND: A combined method for author name disambiguation

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ABSTRACT

Author Name Disambiguation (AND) in digital bibliographic repositories is a persistent challenge due to homonyms and synonyms, compromising information retrieval and database integrity. This work presents a novel framework with a **Com**bined **M**ethod for **A**uthor **N**ame **D**isambiguation (ComMAND) that integrates transfer learning with SciBERT, Graph Convolutional Network (GCN), and Graph-enhanced Hierarchical Agglomerative Clustering (GHAC) to enhance AND performance. The framework includes a Graphical User Interface (GUI), allowing users to load datasets, execute AND tasks, and visualize results without requiring programming knowledge. By semantically analyzing document content and leveraging graph-based relationships, our approach achieves higher precision in identifying unique authors. Experimental results on AMiner-12, AMiner-18, and DBLP validate the effectiveness of the framework. Considering the DBLP dataset, which contains extensive ambiguous name references (679), the results show the highest F1 of 0.869 and K-metric of 0.972 compared to the baseline works, with improvements ranging from 1.1% to 33.6% over baseline works. These findings highlight the effectiveness of combining machine learning, graph-based techniques, and clustering for large-scale AND tasks.

1. Introduction

Digital bibliographic repositories, such as AMiner (2005-2024), CiteSeerX (2007-2019), DBLP (1993-2024), and PubMed Central® (PMC) are crucial for bibliographic citation data, offering tools to identify scientific works, authors, and their academic networks. DBLP[1] currently catalogs over 7.6 million publications in Computer Science, including journal articles, conference proceedings, workshops, monographs, theses, and textbooks. As of December 2024, it includes approximately 3.67 million authors, 6824 conferences, and 1877 journals, showcasing its extensive coverage of scientific contributions and its continuous expansion to reflect the growing volume of academic research (DBLP, 1936–2024). ArnetMiner[2] stores data on more than 303 million scientific works, 60 million authors, and 2.4 billion bibliographic citations (AMiner, 2024).

By housing millions of bibliographic records, these repositories serve as indispensable resources for the global academic community, enabling centralized searches for relevant publications (Ferreira et al., 2020). Beyond search functionality, they provide analysis tools and features that support decision-making for funding agencies and academic institutions (Hussain & Asghar, 2017). However, challenges such as author name ambiguity caused by abbreviations, typographical errors, or shared names pose significant

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[1] https://dblp.org/.
[2] https://www.aminer.org/.

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obstacles to accurate information retrieval (Xiong et al., 2021). Thus, name ambiguity significantly affects the performance of document and information retrieval in Web search engines and undermines the integrity of entities in integrated databases.

Author name ambiguity in digital bibliographic repositories arises when a single author is associated with multiple name variations (synonyms) or when various authors share an identical name or similar name variations (homonyms) (Ferreira et al., 2020; Veloso et al., 2012). Several works addressing the Author Name Disambiguation (AND) problem appear in the literature, exploring various techniques. Although researchers have studied this problem for decades, it still needs a canonical solution (Han et al., 2004; Han, Xu et al., 2005; Han, Zha et al., 2005; Kang et al., 2009; Tang et al., 2012; Torvik & Smalheiser, 2009; Torvik et al., 2005; Vu et al., 2008). Resolving this issue remains a critical research challenge, especially as digital bibliographic repositories evolve to become more person-centric than document-centric (Ferreira & Laender, 2023; Rodrigues et al., 2021; Shin et al., 2014).

Automatic methods based on graphs, machine learning, and heuristics have been applied to the AND problem, highlighting the diversity of approaches available in different contexts (Ferreira et al., 2020). Recent methods based on Large Language Models (LLMs) have been explored in AND tasks. Sancheti et al. (2024) focused on LLM-driven named entity extraction and ChatGPT's contextual comprehension (revalidation) for web profile processing. Zhang et al. (2024) explored LLMs' ability to disambiguate scholars globally. Yan and AsirAsir (2024) and Zhao and Chen (2025) are successful methods in the KDD Cup'24 (Open Academic Graph Challenge) that involved combining LLMs with algorithms for paper relevance and refining with multi-source models.

Despite their promise, current LLM approaches have limitations and need further examination for meaningful improvements in complex AND scenarios. Zhao and Chen (2025), highlight that relying solely on paper metadata restricts LLM performance, demanding complex comparison strategies with fine-tuning, challenges in multilingual contexts, and heterogeneous data. van Lieshout (2024), agrees that more informative attributes increase the chance of an accurate author match, parameter adjustments (e.g., temperature), and different system prompts (e.g., a chain of through prompting) could potentially improve the accuracy of the LLM approach but costly in terms of time. Yan and AsirAsir (2024), reinforce that isolated LLM use is often simplistic and that hybrid approaches combining LLMs with traditional models, like LightGBM, achieve better results. These studies suggest that LLMs alone are not yet sufficient to address the full complexity of the AND task. In summary, effective AND might combine LLM-driven natural language understanding with specialized algorithms, databases, and network analysis techniques to accurately perform complex AND scenarios. In the comprehensive overview of LLMs presented by Naveed et al. (2024), the authors discuss challenges to handling more complex tasks, including computational cost, overfitting, limited knowledge, multimodality, catastrophic forgetting, interpretability, explainability, hallucinations, and prompt engineering, among others. All these factors make current LLM-based solutions, alone, less suitable for large and structurally complex scenarios.

Considering the AND research scenario, this work's objective focuses on developing a framework to enhance disambiguation in digital bibliographic repositories, considering meaningful semantic hierarchical and graphical clusters to boost authors' identification. In a previous work Rodrigues and Ralha (2024), we developed a hybrid machine-learning method to validate the AND approach, aiding the framework definition. The novel framework proposed in this work integrates machine learning techniques, especially transfer learning, Graph Convolutional Networks (GCN), and Graph-enhanced Hierarchical Agglomerative Clustering (GHAC), improving the AND accuracy and effectiveness. Transfer learning for contextual semantic representations enables models to apply knowledge from one domain to another, facilitating tasks with minimal labeled data. Using pre-trained models enhances understanding of article topics and distinguishes between individuals with similar names. A recent work using a pre-trained model for Chinese word sense disambiguation achieves good results with RS-BERT (Zhou, Huang et al., 2024). GCN effectively manages graph-structured data, with a focus on author relationships in AND, as well as connections between authors, publications, and citations. They model co-authorship, citation links, and other network features. By spreading information in the graph, GCN uncovers hidden relationships and cues vital for accurately identifying authors. The use of graph-structure data is also used in Li et al. (2023) for medical abbreviation disambiguation, presenting a framework model (MADRFM).

GHAC combines hierarchical clustering with graph methods to group similar entities based ond their relationships and features. It clusters authors by analyzing citation and co-authorship networks to identify publication clusters associated with the same author. By improving traditional hierarchical clustering with graph data, GHAC forms meaningful clusters representing individual authors or collections of works, helping in better author identification. In summary, transfer learning allows contextual semantic embeddings of publications, which are graph-structured by heterogeneous bibliographic author networks accurately identified by GCN, with publications and authorship tied up through meaningful clusters to boost AND complex tasks.

Thus, this research's main contribution is:

- a novel framework for AND with a **Com**bined **M**ethod for **A**uthor **N**ame **D**isambiguation (ComMAND) and a user-friendly Graphical User Interface (GUI). ComMAND integrates transfer learning with SciBERT to contextual semantic embeddings for documents, directly refined by a GCN leveraging structural information from heterogeneous bibliographic networks, with GHAC clustering to yield meaningful clusters representing distinct works.
- extensive experiments conducted on legacy datasets, AMiner-12, AMiner-18, and DBLP, which differ in size, attributes, and ambiguity levels, to verify the effectiveness of the proposed framework.

The rest of the article includes in Section 2 a formal definition of the used techniques and Section 3 with related work. Section 4 details the novel framework with ComMAND. Section 5 describes the experiments, including the description of the datasets, evaluation metrics, experimental setup, and the ablation studies. Section 6 analyzes the baseline comparative works. Section 7 includes discussion of theoretical and practical implications. Finally, Section 8 concludes the work and outlines future research directions.

2. Preliminaries

In this section, we present the definition for the techniques used in the proposed framework, including the AND task (Definition 1), transfer learning (Definition 2), heterogeneous networks (Definition 3), GCN (Definition 4), and GHAC (Definition 5).

Definition 1 (*AND Task*)**.** As defined by Ferreira et al. (2020), the AND task involves organizing a collection of citation entries, represented as $C = \{c_1, c_2, \dots, c_k\}$. For a given entry $c_i \in C$, each value in its *authornames* attribute corresponds to a unique author and links to an authorship record r_j. The goal of an AND method is to create a disambiguation function that partitions the set of authorship records $\{r_1, r_2, \dots, r_m\}$ into n groups $\{a_1, a_2, \dots, a_n\}$, ensuring each group a_i contains all and only the records associated with a single author.

Definition 2 (*Transfer Learning*)**.** Transfer learning is a machine learning technique where a model developed for a particular task is reused as the starting point for a model on a second related task.

Definition 2 means that:

- Source domain task — the domain and task in which the model is initially trained, typically with a large amount of labeled data.
- Target domain task — the domain and task where the model is applied, which have limited labeled data or differ in some way from the source domain.
- Knowledge transfer — transferring the learned representations or parameters from the source task to the target task to improve performance on the latter.

Typical applications include natural language processing (NLP) and computer vision, where related tasks share data structures. As defined by Weiss et al. (2016), transfer learning in NLP reuses knowledge from one task to improve another. Key models include Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) and the Pretrained Language Model for Scientific Text (SciBERT) (Beltagy et al., 2019), which use self-supervised tasks for learning and fine-tuning. We use co-authorship relationships, publication attributes, and textual embeddings from SciBERT.

Definition 3 (*Heterogeneous Networks*)**.** As defined by Shi et al. (2017), heterogeneous networks are graphs composed of multiple types of entities (nodes) and relations (edges). Formally, a heterogeneous network is a graph $G = (V, E, \mathcal{T}_v, \mathcal{T}_e)$, where V is the set of nodes, E is the set of edges, $\mathcal{T}_v$ denotes the set of node types, and $\mathcal{T}_e$ denotes the set of edge types. Each node $v \in V$ is associated with a type $\phi(v) \in \mathcal{T}_v$, and each edge $e \in E$ has a type $\psi(e) \in \mathcal{T}_e$.

In this work, within the context of bibliographic data, nodes typically represent publications, authors, keywords, or venues, while edges capture relationships such as authorship, co-authorship, shared venues, or shared keywords. Heterogeneous networks facilitate modeling relational structures, such as recommendation and link prediction tasks.

Definition 4 (*Graph Convolutional Networks*)**.** Graph Convolutional Networks (GCNs) are neural architectures designed for learning representations over graph-structured data. They integrate node features and topological information by propagating and aggregating information from local neighborhoods. The original formulation is grounded in spectral graph theory, where graph convolutions are defined in the Fourier domain using the eigendecomposition of the graph Laplacian.

Definition 4 operates on a graph $G = (V, E)$, where V is the set of nodes and E is the set of edges. Each node $v \in V$ has an associated feature vector $\mathbf{x}_v \in \mathbb{R}^F$, where F denotes the number of input features. The graph structure is encoded by an adjacency matrix A, where $A_{ij} = 1$ if there is an edge between nodes i and j, and $A_{ii} = 1$ when self-loops are added.

To reduce the computational cost of spectral convolutions, Kipf and Welling (2017) proposed a first-order approximation, resulting in the following layer-wise propagation rule:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right),$$

where:

- $H^{(0)}$ = is the input feature matrix X,
- $\tilde{A} = A + I$ is the adjacency matrix with added self-loops,
- $\tilde{D}$ is the degree matrix of $\tilde{A}$,
- $W^{(l)}$ is the trainable weight matrix at layer l,
- $\sigma(\cdot)$ is a nonlinear activation function, such as ReLU.

This formulation enables localized feature aggregation from immediate neighbors, effectively capturing structural patterns in the graph. After several layers, the final output $H^{(L)}$ encodes rich node representations suitable for downstream tasks such as classification or clustering. Due to its scalability and ability to exploit both feature and structural information, GCN has become a widely adopted method in semi-supervised learning over graphs.

Definition 5 (*Graph-enhanced Hierarchical Agglomerative Clustering*)**.** As defined by Qiao et al. (2019), GHAC is a method that integrates the topological information of heterogeneous graphs into the Hierarchical Agglomerative Clustering (HAC) process.

Initially, each document is considered as a single cluster. In each iteration, the algorithm merges the pair of clusters with the highest similarity based on the average cosine similarity between document embeddings.

Definition 5 operates on a heterogeneous graph $G = (V, E)$, where V is the set of nodes representing entities such as publications, authors, or venues, and $E \subseteq V \times V$ is the set of edges representing relationships between these entities. Each document node $v \in V$ is associated with an embedding vector $z_v \in R^d$. The similarity between two clusters C_i and C_j is computed using the average pairwise cosine similarity:

$$\mathrm{sim}(C_i, C_j) = \frac{1}{|C_i||C_j|} \sum_{u \in C_i} \sum_{v \in C_j} \frac{z_u^\top z_v}{\|z_u\| \cdot \|z_v\|}.$$

The algorithm iteratively merges the most similar clusters until a stopping criterion is met. When the number of expected clusters K is known – for example, based on the number of true authors – the algorithm stops when exactly K clusters have been formed. In scenarios where K is not known in advance, the number of clusters can be determined automatically by evaluating clustering quality criteria, such as modularity.

3. Related work

To guide the discussion and analysis of related work, we sought to analyze literature reviews already carried out in AND, published between 2012 and 2024. These reviews have varied focuses, including the creation of taxonomies and the classification of automatic AND methods (Ferreira et al., 2012), the analysis of AND techniques applied to the Pubmed bibliographic repository (Sanyal et al., 2021), the identification of automatic AND approaches in bibliographic repositories (Ferreira et al., 2020), and the presentation of established techniques alongside current research frontiers in the AND domain (Rodrigues et al., 2024).

As presented in the literature review of Rodrigues et al. (2024), approaches based on graphs, word embedding with supervised learning, and heuristics with probabilistic applications are widely used to solve the AND problem. The literature review highlights author clustering techniques as prevalent and effective methods. The review presents approaches to AND related to our solution, including new references that use graph-based solutions, GCN, and transfer learning. Following the categorization proposed in this review, we focus on works primarily classified as author grouping with similarity functions and author assignment with clustering. The review also indicates that most methods employ AMiner and DBLP as bibliographic bases to extract information. The following sections discuss our AND literature review.

3.1. Graph embedding approaches

The approach proposed in Pooja et al. (2020), called ATGEP, addresses the AND problem by creating two graphs for each ambiguous reference: an author similarity graph based on coauthors' overlap between documents and a topic similarity graph. The approach eliminates conflicting connections from the topic graph based on the similarity of references. Then, combine this graph with the author's similarity graph to identify different clusters. Although this method does not use representation learning for AND, it considers meaningful relationships between documents.

Rastogi et al. (2023) investigate graph-based methods for AND using bipartite graphs of papers and authors. They evaluate random walk-based similarity, Node2Vec embeddings, and Graph Neural Networks (GNNs) in supervised and unsupervised settings. Features such as titles, abstracts, and organizations enrich node embeddings. Random walks achieve high precision but low recall, while embedding-based methods improve recall but may group unrelated profiles.

Ma et al. (2020) presents Mech-RL, a module for AND in heterogeneous networks in the context of the Internet of Things. The method combines relational and textual data with meta-paths and graph embeddings to perform hierarchical clustering and disambiguation. The framework efficiently integrates heterogeneous networks and textual information, outperforming existing methods in complex bibliographic databases.

Wang et al. (2020) presents an approach for AND that uses heterogeneous networks and Node2Vec to generate node embeddings, capturing structural and semantic relationships between authors, documents, and attributes. Adversarial learning refines these representations, making them more robust to variations and noise in the data. According to the authors, the approach stands out for exploring high-order connections and improving accuracy in the AND task in complex scenarios.

Graph embedding approaches for AND use structural and semantic relationships in heterogeneous networks to create node representations. However, ATGEP (Pooja et al., 2020) lacks adaptability, Node2Vec methods focus on local patterns but overlook broader contexts (Rastogi et al., 2023; Wang et al., 2020), and Mech-RL's reliance on predefined meta-paths restricts its generalization across various networks (Ma et al., 2020). In contrast, our proposed framework captures local interactions and broader relational patterns without relying on predefined pathways, using heterogeneous graphs integrating a wider range of structural and contextual document information, achieving the effectiveness of the AND task.

3.2. Transfer learning approaches

Kim and Owen-Smith (2020) explore transfer learning on AND tasks where no labeled data was available for training. The results showed that AND models developed through transfer learning can perform effectively. When training source data represent the main

characteristics of the target datasets, these models produce results comparable to those performed by traditional machine learning approaches. Tradition methods, in contrast, depend on algorithms trained on specifically labeled subsets of the target data.

Boukhers and Asundi (2023) propose a method for AND using coauthor and research domain information. They preprocess DBLP data by cleaning and tokenizing the text, then transforming it into numerical representations. They encode coauthors with character-level embeddings and titles with Bidirectional Encoder Representations from Transformer (BERT). A neural network learns the relationship between these representations and the target author, enabling AND for new publications.

Huang et al. (2024) present a cross-dataset methodology for AND, focusing on transfer learning and cross-domain adaptation. The approach utilizes co-authorship relationships, publication attributes, and textual embeddings generated by SciBERT. By combining structural and textual features, the proposed framework demonstrates good generalization adaptation across multiple domains, and experimental results show that the proposed model outperforms current state-of-the-art methods.

Transfer learning for AND uses pre-trained language models to comprehend textual data, but approaches vary in their combination with heterogeneous graph features. Some methods limit generalization by assuming similar datasets (Kim & Owen-Smith, 2020), while others utilize BERT and SciBERT to separate textual and structural inputs (Boukhers & Asundi, 2023; Huang et al., 2024). In contrast, our approach seeks to integrate structural and semantic information more directly. By combining pre-trained language model embeddings with graph-based representations, the model can capture diverse relationships and improve the effectiveness of the AND task in datasets with complex textual and structural patterns.

3.3. GCN approaches

Zhou, Shi et al. (2024) propose an AND framework focused on large-scale bibliographic datasets. The framework combines a natural language processing model for semantic feature extraction with an unsupervised GCN-based approach for relationship and graph feature learning. The experiments and validation used the AMiner-18 dataset. The framework demonstrates strong performance on real-world datasets characterized by the AND problem while addressing scalability and semantic diversity.

Zhou, Chen et al. (2024) presents a hybrid model that integrates graph attention mechanisms for AND. The approach combines raw feature embeddings with graph-based representations, using attention layers to capture local and global structural information. The authors constructed a new dataset, AD-AND, and validated their method on the OAG-WhoisWho dataset (Chen et al., 2023). According to the authors, their approach presents precision and recall improvements compared to the baselines.

Choi et al. (2024) propose an approach for automatic AND in academic search engines. The proposed method uses a rule-based approach and deep learning techniques, incorporating metadata such as author names, affiliations, coauthors, and publication attributes. The approach also uses clustering algorithms, such as HAC, and a multi-classifier framework to adapt to variable metadata characteristics. According to the authors, this methodology achieved good performance in terms of F1 score and demonstrates robustness in heterogeneous datasets by normalizing ambiguous names and integrating multiple metadata sources.

Pooja et al. (2021) propose an unsupervised learning method for AND that tackles issues like homonyms, synonyms, and sparsely labeled data. The approach uses relational (coauthorship) and non-relational (title, abstract, venue) features to build coauthor and meta-content similarity graphs. Word2Vec transforms these attributes into vector representations, which variational graph auto-encoders use to generate low-dimensional embeddings. Neural networks fuse the embeddings, and hierarchical clustering assigns authorship.

Pooja et al. (2022) propose an AND approach that combines GCN with attention mechanisms to learn representations in a heterogeneous document graph. The method emphasizes the use of attention to capture relevant context from different types of neighbors and relationships, enhancing semantic and contextual information. The authors evaluate their approach on two variants of the ArnetMiner dataset, AMiner-12 and AMiner-18, highlighting the impact of attention mechanisms on node representation learning.

GCN-based methods for AND merge structural and semantic information from various graphs, differing in integrating text and managing complexity. Some emphasize lightweight designs (Zhou, Shi et al., 2024) while others use attention mechanisms (Zhou, Chen et al., 2024) and heterogeneous metadata but process textual and structural data separately, often missing their interplay (Choi et al., 2024; Pooja et al., 2021, 2022), each with trade-offs in cost and data handling. Our proposed framework integrates pre-trained language model embeddings into the network graph, enabling joint learning of structural and semantic signals without relying on handcrafted features or treating text and structure as disjoint inputs.

Table 1 presents an overview of the related work compared to the framework with ComMAND presented in this work. Note that our approach differs since it uses graph embeddings, transfer learning, GCN, and GHAC, validated with AMiner-12, AMiner-18, and DBLP datasets, with pairwise F1 and K-metrics metrics as the most used ones considering the related work.

4. Framework

Considering the absence of a canonical solution and the open AND problem in digital bibliographic repositories, we present a novel framework with ComMAND that integrates transfer learning, GCN, and GHAC. The framework code and sample datasets used in our experiments are available in the repository.[3] Transfer learning with SciBERT captures contextual semantics from texts and can outperform traditional embeddings such as TF-IDF and Word2Vec. The GCN efficiently models relationships in heterogeneous

[3] https://github.com/natansr/ComMAND.git.

Table 1
Related work comparison overview.

Reference	Dataset	Evaluation metric	Method			
			Graph embeddings	Transfer learning	GCN	GHAC
Zhou, Shi et al. (2024)	Aminer-18	Pairwise F1			✓	
Zhou, Chen et al. (2024)	OAG-WhoisWho, AD-AND	Pairwise Micro/Macro F1			✓	
Choi et al. (2024)	ScienceOn, KCI, NTIS	Precision F1-score			✓	
Pooja et al. (2021)	AMiner-12, AMiner-18	Pairwise F1 K-Metric BF1			✓	
Pooja et al. (2022)	AMiner-12, AMiner-18, DBLP	Pairwise F1 K-Metric			✓	
Kim and Owen-Smith (2020)	Synthetic	F1-Score B-Cubed		✓		
Boukhers and Asundi (2023)	Academic-II, AMiner-AND	Micro/Macro F1		✓		
Huang et al. (2024)	Academic-II, AMiner-AND	F1-score		✓		
Pooja et al. (2020)	AMiner-12, AMiner-18, DBLP	Pairwise F1 K-Metric	✓			
Rastogi et al. (2023)	Synthetic	Pairwise F1	✓			
Ma et al. (2020)	Synthetic	F1-score	✓			
Wang et al. (2020)	AceKG	F1-score	✓			
Framework with ComMAND	AMiner-12, AMiner-18, DBLP	Pairwise F1 K-Metric	✓	✓	✓	✓

networks by integrating textual and structural information, which stands out from conventional Convolutional Neural Networks (CNNs). The GHAC, a graph-based clustering technique, improves clustering by leveraging the graph structure, ensuring greater cohesion and context than conventional methods, such as K-means or DBSCAN. By combining these techniques, ComMAND enhances the accuracy and scalability of AND tasks. The quantitative evaluation uses F-measure (Christen et al., 2023; Ferreira et al., 2020) and K-metric (Corregidor & Martínez-Pérez, 2021; Estrada-Moreno et al., 2014; Jannesari & Omoomi, 2012; Santana et al., 2017), enabling comparison of its performance against literature state-of-the-art approaches.

4.1. Architecture

This section aims to establish a structure defining the operations in each architecture layer, from the initial data input to the resulting output after executing the algorithm. Fig. 1 illustrates the framework with ComMAND architecture with four layers: Layer 1 (L_1), Layer 2 (L_2), Layer 3 (L_3), and Layer 4 (L_4).

4.1.1. Data entry and preprocessing

The L_1 is responsible for data entry and processing. Fig. 1 presents in L_1 a component that allows user access, bibliographic repositories, system elements, and also the inclusion of new datasets. These datasets can contain bibliographic information relevant to the AND task in digital bibliographic repositories. The user access in L_1 has a GUI, which receives data for the AND task. Upon the complete dataset submission, L_1 executes a preprocessing pipeline to ensure data consistency. This pipeline includes deduplication of publication entries, normalization of textual fields (e.g., removal of special characters/symbols and excess whitespace), and standardization of metadata fields (e.g., ensuring the presence of required fields) to ensure data readiness for graph construction and embedding generation. Once the dataset is provided and processed, the interface forwards these data to the L_2. The GUI is also responsible for making the final disambiguation results generated by the framework available to users. In addition to handling data input and result visualization, the GUI plays a central role in improving the framework's usability for non-expert users, lowering the barrier to entry for applying AND techniques in real-world bibliographic management and digital libraries.

4.1.2. Heterogeneous graph creation and embeddings

The L_2 of the framework plays a fundamental role in creating the structure of the heterogeneous network from the information received from L_1 and providing contextual information for the AND task. Thus, we have a branch of the information obtained from the documents inserted in L_1. One branch involves the creation of a heterogeneous graph using the document relations that include different types of nodes and edges. Following the concept introduced in Definition 3, the heterogeneous graph is defined as:

$$G_{\text{heterogeneous}} = (N_{\text{nodes}}, E_{\text{edges}}),$$

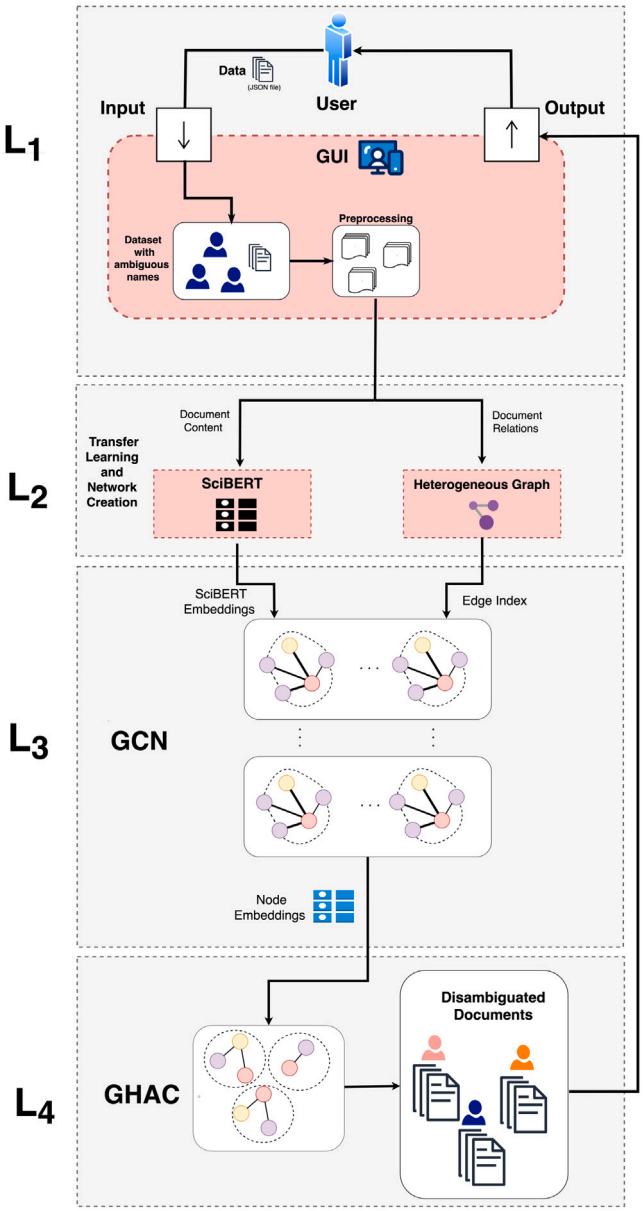

Fig. 1. The framework with ComMAND layered architecture.

where:

- N_{nodes} includes different types of nodes, such as publications, authors, venues, abstracts, and keywords, representing the entities in the graph.
- E_{edges} captures the various connections between nodes of the $G_{\text{heterogeneous}}$, including:
  - *contains*: linking publications to keywords.
  - *written_by*: connecting publications to authors.
  - *co_authored*: linking authors who collaborate in publication.
  - *has_abstract*: linking publications to their corresponding abstract nodes.
  - *shared_word*: connecting publications sharing common keywords.
  - *published_in*: linking publications to their venues (e.g., conferences or journals).

The other branch in L_2 calculates embeddings for publications using SciBERT's contents based on titles, abstracts, and keywords. This process aligns with the principles of transfer learning outlined in Definition 2, where a language model pre-trained on scientific text is adapted to provide semantic representations for AND task (Definition 1). The method assigns these embeddings with a vector of numbers representing contextual words as features to the nodes representing publications in the graph.

Considering a set of N publications with titles, abstracts, and keywords, where each document i has a title T_i, an abstract R_i, and keywords K_i, the process for extracting and calculating embeddings includes three steps:

1. Tokenization: tokenize each title T_i, abstract R_i, and keyword set K_i into sequences of individual tokens, represented as $\{t_{i,1}, t_{i,2}, \dots, t_{i,L_i}\}$, $\{r_{i,1}, r_{i,2}, \dots, r_{i,M_i}\}$, and $\{k_{i,1}, k_{i,2}, \dots, k_{i,P_i}\}$, respectively, where L_i, M_i, and P_i denote the token counts for the title, abstract, and keywords.
2. SciBERT embedding generation: process the token sequences of titles T_i, abstracts R_i, and keywords K_i with SciBERT. Combine pre-trained language models with dimensionality reduction techniques to produce a vector of embeddings for each publication. These embeddings capture the semantics of the texts, reflecting the main topics and contextual relationships.
3. Graph integration: use the embeddings generated by SciBERT as features of the nodes representing publications in the heterogeneous graph.

The edge index represents the connections between nodes, avoiding the need to generate a dense adjacency matrix. This format enables the GCN to handle large heterogeneous networks while preserving essential connectivity information between the various entities.

4.1.3. GCN learning

The L_3 (Fig. 1) corresponds to the GCN model (Definition 4). It operates on the heterogeneous graph $G = (V, E)$, where each node $v \in V$ is represented by a feature vector $x_v \in R^F$. Document nodes receive SciBERT embeddings derived from their titles, abstracts, and keywords. The remaining nodes, such as authors and venues, are initialized with zero vectors, contributing only structurally to the learning process.

The node features compose a matrix $X \in R^{|V|\times F}$, where each row corresponds to a node in the graph. This matrix X serves as the input $H^{(0)}$ to the GCN. The network performs layer-wise propagation according to the rule of Definition 4, using the normalized adjacency matrix $\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}$ to aggregate information from neighboring nodes. At each layer l, the node features are updated via a trainable weight matrix $W^{(l)}$ and passed through a nonlinear activation function $\sigma(\cdot)$ (such as ReLU). After L layers, the final output $H^{(L)} \in R^{|V|\times F'}$ contains low-dimensional embeddings for all nodes. These embeddings encode the initial node features and the structural context captured by the graph topology.

To train the model, we apply a reconstruction loss function Mean Squared Error (MSE) that encourages the final embeddings Z_i (i.e., the ith row of $H^{(L)}$) to remain close to the original input features x_i. The MSE loss function is defined by:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(Z_i - X_i)^2,$$

where N is the number of nodes, Z_i represents the GCN output for node i, and X_i denotes the original feature vector of node i. The Adam optimizer adjusts the model weights with an appropriately tuned learning rate (Kingma & Ba, 2015).

Once trained, the GCN generates low-dimensional node embeddings that capture both the nodes' initial features and the structural relationships in the graph. These embeddings serve as input for subsequent tasks, such as clustering with GHAC.

After training, the GCN generates low-dimensional embeddings for all nodes in the graph. During the clustering phase with GHAC, we use only the embeddings of document nodes. Algorithm 1 outlines the GCN training procedure and takes the number of layers L, the number of training epochs T, heterogeneous graph $G = (V, E)$, document embeddings $X_{\text{doc}} \in \mathbb{R}^F$ as input, where F denotes the number of features. Users should define these parameters empirically, considering the dataset's characteristics and the model's convergence behavior. In our experiments, we examined how different values of L affect the AND task performance, as presented in Section 5.3.2. We selected a fixed value for T based on preliminary runs that consistently reached convergence, using loss monitoring and checkpoint validation to guide the process.

The training computational complexity described in Algorithm 1 grows with the number of epochs T, layers L, edges $|E|$, and input feature dimension F. Each layer applies a graph convolution defined by the rule in Definition 4, which involves matrix operations over the normalized adjacency structure. This computation requires $\mathcal{O}(|E|F)$ per layer. The loss function introduces an additional cost of $\mathcal{O}(|V|F)$ per epoch. As a result, the overall complexity of training the GCN reaches $\mathcal{O}(T \cdot L \cdot |E| \cdot F)$.

4.1.4. Hierarchical agglomerative clustering

The L_4 (Fig. 1) receives as input the final node representations $H^{(L)} \in \mathbb{R}^{|V|\times d}$ generated by the GCN, as described in Section 4.1.3. From this matrix, we extract the embedding vectors $\{\mathbf{z}_i\}_{i=1}^{n} \subseteq \mathbb{R}^d$, corresponding only to the publication nodes. These embeddings serve as input to the clustering process.

This step applies the GHAC technique (Definition 5). GHAC begins by treating each publication as an individual cluster $C_i = \{i\}$. At each iteration, it computes the similarity between every pair of clusters based on the average cosine similarity of their constituent embeddings. There is a merge of the two most similar clusters, with the similarities updated accordingly.

Algorithm 1: GCN training of heterogeneous graph.

Input: Heterogeneous graph $G = (V, E)$, document embeddings X_{doc}, number of layers L, number of epochs T
Output: Refined node embeddings Z
Step 1: Prepare node features
foreach *node* $v \in V$ **do**
  if v *is a document* **then**
    Set $x_v \leftarrow X_{\text{doc}}[v]$
  else
    Set $x_v \leftarrow 0$
Construct feature matrix X from all x_v
Construct edge index from E
Step 2: Initialize GCN
Initialize L-layer GCN with weights $\{W^{(1)}, \dots, W^{(L)}\}$
Step 3: Train GCN
for $t = 1$ **to** T **do**
  $H^{(0)} \leftarrow X$ **for** $\ell = 1$ **to** L **do**
    $H^{(\ell)} \leftarrow \text{ReLU}\left(\text{GCNConv}(H^{(\ell-1)}, E; W^{(\ell)})\right)$
  $Z \leftarrow H^{(L)}$
  Compute loss: $L \leftarrow \frac{1}{|V|} \sum_{v \in V} \|Z_v - x_v\|^2$
  Update weights using Adam optimizer
return Z

The merging continues until the number of clusters equals a predefined value K, corresponding to the number of distinct authors associated with an ambiguous name (i.e., given by the ground truth). Although the author labels untaken during clustering, the value of K is assumed to be known and guides the termination criterion. Algorithm 2 outlines the GHAC procedure.

Algorithm 2: GHAC.

Input: Set of publication embeddings $\{z_i\}_{i=1}^n$, number of clusters K
Output: Cluster labels $\{y_i\}_{i=1}^n$
Step 1: Initialization
Create initial clusters $\mathcal{C} \leftarrow \{C_1, C_2, \dots, C_n\}$, where each $C_i = \{i\}$
Step 2: Compute initial similarities
foreach *pair of clusters* (C_i, C_j) **do**
  Compute average cosine similarity:

$$\text{sim}(C_i, C_j) = \frac{1}{|C_i||C_j|} \sum_{u \in C_i} \sum_{v \in C_j} \frac{z_u^\top z_v}{\|z_u\| \cdot \|z_v\|}$$

Step 3: Iterative merging
while $|\mathcal{C}| > K$ **do**
  Find $(C_p, C_q) = \arg\max_{(C_i, C_j)} \text{sim}(C_i, C_j)$
  Merge clusters: $C_{\text{new}} \leftarrow C_p \cup C_q$
  Update $\mathcal{C} \leftarrow \mathcal{C} \setminus \{C_p, C_q\} \cup \{C_{\text{new}}\}$
  Recompute similarities between C_{new} and all other clusters
Step 4: Output
Assign cluster label y_i to each document i based on final cluster membership
return $\{y_i\}_{i=1}^n$

The Algorithm 2 follows the same HAC procedure initially proposed by Qiao et al. (2019), but we compute similarities differently. Instead of reconstructing a sparse graph and comparing only connected nodes, we calculate cosine similarity between all pairs of publication embeddings $\{z_i\}_{i=1}^n$, where n is the number of documents in a disambiguation block and each $z_i \in \mathbb{R}^d$ is a vector of dimension d. This approach produces a full similarity matrix through $\binom{n}{2}$ pairwise comparisons, each with cost $\mathcal{O}(d)$, resulting in a total computational complexity of $\mathcal{O}(n^2 d)$ for the initial computation. We then iteratively merge the most similar clusters until only K remain. Although this strategy incurs a higher computational cost than the sparse-graph-based GHAC variant, it simplifies the pipeline by removing the need for graph reconstruction and integrating directly with embedding-based representations.

The clustering process leverages *Python* libraries such as *NetworkX*, *Community*, and *Sklearn* to handle large networks efficiently. The method enhances internal cluster cohesion while preserving semantic and structural relationships, effectively organizing academic networks , and revealing patterns of co-authorship and collaboration. After clustering, the technique assigns publications to specific authors.

Table 2
Comparison of attributes across AMiner-12, AMiner-18, and DBLP datasets.

Attributes	AMiner-12	AMiner-18	DBLP
Ambiguous name references	109	100	679
Publications	7447	70,258	6478
Unique authors	1546	12,798	1463
Title	✓	✓	✓
Abstract		✓	✓
Keywords			✓
Coauthors	✓	✓	✓
Organization	✓		
Venue	✓	✓	✓
Manual curation	✓	✓	✓

Finally, the evaluation compares predicted labels to ground truth labels, each identifying a specific author. The process averages metrics across all evaluated publications to measure clustering performance. Section 5.2 details the validation methods and metric calculations.

5. Experiments

The experiment's goal was to validate the proposed framework, particularly the contribution of combining semantic embeddings generated by SciBERT with graph-based learning using GCN, followed by clustering with GHAC. This evaluation demonstrates the impact of each component on the effectiveness of AND across datasets.

5.1. Datasets

In this work, we used AMiner-12, AMiner-18, and DBLP datasets, which differ in terms of size, attributes, and levels of ambiguity. As verified in the literature review in Rodrigues et al. (2024), these datasets are widely used and recognized as benchmarks in AND tasks. They are considered legacy datasets providing diverse challenges and variations for evaluating AND techniques. Furthermore, their manual curation ensures a high standard of data quality, making them reliable references for evaluating AND models.

As shown in Table 2, AMiner-12 dataset[4] comprises 109 ambiguous name references, covering 7447 publications from 1546 unique authors, including title, coauthors, organization, and venue attributes (Tang et al., 2012). The AMiner-18 dataset[5] contains 100 ambiguous name references, covering 70,258 publications and 12,798 unique authors (Zhang et al., 2018). Compared to AMiner-12, it includes additional features such as abstracts, offering richer metadata for analysis. The DBLP dataset created by Qian et al. (2015)[6] includes 679 ambiguous name references, covering 6478 publications and 1463 unique authors with title, abstract, keywords, venues, and coauthors attributes.

Although the AMiner-12 dataset includes the affiliation attribute (organization), this feature is not consistently available across all datasets (i.e., AMiner-18 and DBLP). In contrast, only DBLP includes keyword annotations that are unavailable in the AMiner-12 and AMiner-18 datasets.

5.2. Evaluation metrics

To evaluate the framework performance on AND problems, we use two main metrics, pairwise F1 (pF1) and the K-metric. These capture different aspects of the quality of clustering and document assignment to authors. The pF1 metric is widely used in entity resolution problems calculating from pairwise Precision (pP) and Recall (pR) (Ferreira et al., 2020).

The pP measures the proportion of correctly classified documents to the total number of classified documents. It evaluates the ability of the algorithm to associate documents with the correct authors:

$$pP = \frac{\text{documents correctly classified}}{\text{total documents classified}}.$$

The pR evaluates the ability of the algorithm to retrieve all documents that belong to an author, that is, how many documents from a real author the algorithm was able to identify:

$$pR = \frac{\text{documents correctly retrieved}}{\text{total documents from real author}}.$$

The pF1 is the harmonic mean between pP and pR, providing a balanced measure between these two aspects:

$$pF1 = \frac{2 \times pP \times pR}{pP + pR}.$$

[4] https://www.aminer.cn/disambiguation.
[5] https://github.com/neozhangthe1/disambiguation.
[6] https://github.com/yaya213/DBLP-Name-Disambiguation-Dataset.

The K-Metric is a geometric mean (K) composite metric that combines cluster purity using Average Cluster Purity (ACP) and Average Author Purity (AAP) to capture both the quality of clustering and the completeness of document attribution to authors (Santana et al., 2017).

The ACP assesses the purity of predicted clusters relative to real clusters by measuring the proportion of correctly clustered documents into clusters representative of the authors, calculated as:

$$ACP = \frac{1}{N} \sum_{i=1}^{e} \sum_{j=1}^{t} \frac{n_{ij}^2}{n_i},$$

where N is the total number of publication records in the test set, t is the number of real clusters, e is the number of predicted clusters, n_{ij} is the number of elements shared between predicted cluster i and real cluster j, and n_i is the total number of elements in predicted cluster i.

The AAP measures how much the predicted clusters fragment the documents of a real author, evaluating the documents' cohesion belonging to the same author within the clusters:

$$AAP = \frac{1}{N} \sum_{j=1}^{t} \sum_{i=1}^{e} \frac{n_{ij}^2}{n_j},$$

where n_j is the total number of elements in real cluster j.

The geometric mean K of ACP and AAP represents a global measure of clustering quality, considering both the purity of clusters and the correct attribution of documents to authors:

$$K = \sqrt{ACP \cdot AAP}.$$

The pF1 and K-metric provide a comprehensive assessment of the quality of our AND combined method, considering both the accuracy of document assignment to authors and the purity and cohesion of the generated clusters.

5.3. Experimental setup

This section presents the experimental setup. We evaluate the overall performance of the proposed framework in solving the AND task. The evaluation considers the ability to group publications by their authentic authorship, assessed using pF1 and K-Metric. In addition, an ablation study checked the impact of different components on the model's overall performance.

5.3.1. Embedding extraction setup

SciBERT generates vector representations of scientific documents by extracting embeddings from titles, abstracts, and keywords, and then combining these components to produce embeddings, creating 768-dimensional representations for each document. Each document is represented by up to 20 tokens.

To formalize this process, a publication p_i, consisting of a title, abstract, and keywords, is concatenated into a single textual sequence. For example, a document with the title *"Author Name Disambiguation"* and the abstract *"A method to solve AND in bibliographic datasets"* results in the sequence:

p_i = "Author Name Disambiguation. A method to solve AND in bibliographic datasets."

This sequence is tokenized into up to 20 tokens:

$$\text{Tokens}(p_{i,j}) = \{t_{i,1}, t_{i,2}, \ldots, t_{i,20}\},$$

where i represents the publication, and j indicates the position of each token within the token set.

If the sequence contains fewer than 20 tokens (as in the example above), padding tokens fill the remaining slots to meet the limit. For sequences exceeding 20 tokens, the process retains only the initial portion, ensuring a uniform input size for efficient processing.

SciBERT processes each token $t_{i,j}$ and generates an embedding $\mathbf{e}_{i,j} \in \mathbb{R}^{768}$. The final representation of the publication is calculated by aggregating the embeddings of all tokens using the mean:

$$\mathbf{v}_i = \frac{1}{20} \sum_{j=1}^{20} \mathbf{e}_{i,j}, \quad \mathbf{v}_i \in \mathbb{R}^{768}.$$

This vector representation $\mathbf{v}_i$ captures the publication's semantic and contextual information, facilitating further analysis and integration into graph-based models.

5.3.2. GCN configuration

The information and representations from constructing the heterogeneous graph and the SciBERT embeddings were used as initial data to train the GCN. Before defining the final GCN configuration, we conducted a study to verify the ideal number of

Table 3
Comparison of GCN layers using AMiner-12, AMiner-18, and DBLP datasets.

Dataset	Layers	pP	pR	pF1	ACP	AAP	K-Metric
AMiner-12	1	0.782	0.722	0.739	0.913	0.930	0.921
	2	0.791	0.728	0.747	**0.918**	0.932	**0.925**
	3	**0.799**	0.733	0.751	0.913	**0.936**	0.924
	4	0.792	**0.738**	**0.752**	0.909	0.930	0.919
AMiner-18	1	0.776	**0.592**	**0.639**	**0.873**	0.910	0.891
	2	**0.778**	0.587	0.636	0.871	**0.920**	**0.895**
	3	0.709	0.520	0.570	0.840	0.896	0.867
	4	0.701	0.492	0.547	0.829	0.882	0.854
DBLP	1	0.860	0.873	0.864	0.969	0.972	0.970
	2	**0.863**	**0.880**	**0.869**	**0.970**	**0.973**	**0.972**
	3	0.858	0.877	0.865	0.970	0.972	0.971
	4	0.862	0.875	0.865	0.970	0.972	0.971

GCN layers, ranging from one to four, to understand network depth's impact on validation metrics. This study used three datasets, AMiner-12, AMiner-18, and DBLP, as mentioned in Section 5.1. The results are shown in Table 3, revealing consistent tendencies in performance metrics up to three layers, generally providing the best balance.

For the AMiner-12 dataset, the performance differences across layers are less pronounced compared to the other datasets. The highest pR and pF1 values occur at four layers, with a marginal improvement of 0.13% over three layers and 0.67% over two layers. Similarly, pP peaks at three layers, showing an improvement of 1.01% over two layers. Clustering metrics present mixed results: ACP and K-Metric perform best at two layers, while AAP reaches its maximum at three layers. These results suggest that the dataset's relatively smaller size, with a simple structure, does not allow deeper GCNs to provide significant advantages, with up to three layers offering comparable performance across metrics.

In the case of AMiner-18, the larger dataset size results in different behavior. Metrics such as pR, pF1, and ACP reach their highest values with one layer, while pP, AAP, and K-metric peak at two layers. However, significant declines occur as the number of layers increases, particularly between levels 2 and 4. For instance, pP drops by 9.89% and pR by 16.19%, highlighting the limitations of deeper GCNs in handling the larger graph structure of AMiner-18, where over-smoothing impacts node distinction.

For the DBLP dataset, two layers yield the best overall results, with pP and pF1 improving by 0.35% and 0.57%, respectively, compared to one layer. Clustering metrics (ACP, AAP, and K-metric) also peak at two layers, confirming that this configuration provides the best balance between performance and complexity.

Thus, this study shows that the optimal number of GCN layers depends on the size and structure of the dataset. For smaller datasets , such as AMiner-12, slight performance differences across layers make it unclear which configuration is most advantageous. For larger datasets such as AMiner-18 and DBLP, two layers are more effective, as additional layers lead to a loss of node distinction due to higher graph complexity. Adjusting the number of GCN layers to the scale and characteristics of the data is crucial to achieving the best performance while avoiding unnecessary complexity.

Apart from the GCN layers configuration, there is an embedding size of 128 using the ReLU activation function ($\sigma(x) = \max(0, x)$). The loss function chosen was the *Mean Squared Error* (MSE), optimized by the Adam algorithm with a 0.0005 learning rate. The model was trained for 30,000 epochs with a batch size of 128. Python language executed the experiments in a Google Colab L4 environment with the hardware accelerator L4, GPU with 22.5 GB of RAM, CPU with 53 GB of RAM, 201.2 GB disk, and the runtime type configured for Python 3. The implemented saving and recovering checkpoints functionality ensures the experiments' continuity during GCN training. Checkpoints are saved every 1000 steps to allow the training to summarize from the last saved point in case of interruption.

5.4. Ablation study

As the framework with ComMAND uses machine learning techniques that can be computationally expensive, we performed an ablation study to evaluate each component's contribution. The model variations were created by removing or modifying specific components, such as the embedding extraction method and GCN. The pF1 and K-metric used to evaluate the results are maintained. The variants are as follows:

- ComMAND complete model – all components with the heterogeneous graph creation and SciBERT embeddings (Section 4.1.2), GCN learning with three layers (Section 4.1.3), and GHAC (Section 4.1.4).
- Ablation 1 - ComMAND without GCN – without the GCN, the clustering is performed directly on document embeddings using GHAC methods. This variant evaluates the impact of using GCN on the quality of disambiguation.
- Ablation 2 - ComMAND with alternative embedding methods – this variant uses instead of SciBERT other embedding extractions with Word2Vec and TF-IDF. The objective is to examine the effectiveness of such embeddings in the semantic representation of documents.

Table 4
Comparison of results with and without GCN across AMiner-12, AMiner-18, and DBLP datasets.

Metric	AMiner-12			AMiner-18			DBLP		
	Without GCN	With GCN	Gain (%)	Without GCN	With GCN	Gain (%)	Without GCN	With GCN	Gain (%)
pP	0.484	**0.799**	+65.08%	0.399	**0.778**	+94.99%	0.7905	**0.863**	+9.18%
pR	0.455	**0.733**	+61.10%	0.354	**0.587**	+65.82%	0.8236	**0.880**	+6.85%
pF1	0.445	**0.751**	+68.88%	0.358	**0.636**	+77.65%	0.7993	**0.869**	+8.73%
ACP	**0.925**	0.913	−1.30%	0.811	**0.871**	+7.40%	0.9539	**0.970**	+1.69%
AAP	0.845	**0.936**	+10.77%	0.896	**0.920**	+2.68%	0.9538	**0.973**	+2.01%
K-metric	0.883	**0.924**	+4.64%	0.852	**0.895**	+5.05%	0.9536	**0.972**	+1.92%

Table 5
Comparison of embedding methods (SciBERT, TF-IDF, and Word2Vec) across AMiner-12, AMiner-18, and DBLP datasets.

Metric	AMiner-12			AMiner-18			DBLP		
	TF-IDF	Word2Vec	SciBERT	TF-IDF	Word2Vec	SciBERT	TF-IDF	Word2Vec	SciBERT
pP	0.744	0.707	**0.799**	0.534	0.463	**0.778**	0.840	0.775	**0.863**
pR	0.689	0.466	**0.733**	0.320	0.407	**0.587**	0.852	0.769	**0.880**
pF1	0.699	0.539	**0.751**	0.368	0.410	**0.636**	0.842	0.767	**0.869**
ACP	**0.918**	0.830	0.913	0.760	**0.867**	0.871	0.959	0.930	**0.970**
AAP	0.930	0.869	**0.936**	0.857	**0.927**	0.920	0.963	0.927	**0.973**
K-metric	**0.924**	0.848	**0.924**	0.806	**0.897**	0.895	0.961	0.928	**0.972**

5.4.1. Ablation 1

When comparing the framework with ComMAND focusing on the GCN use, we observe significant improvements in the evaluation metrics across all datasets, as shown in Table 4.

Incorporating GCN guided substantial improvements in AMiner-12. The pP metric increased by approximately 65%, pR 61%, and pF1 69%. While ACP presents a slight decrease of 1.3%, the AAP metric improved by 11%, and K-metric by 5%, indicating a data fragmentation reduction and better clustering quality. In AMiner-18, the gains were even more pronounced, with pP, pR, and pF1 improving by nearly 95%, 66%, and 78%, respectively, along with increases of 7% in ACP, 3% in AAP, and 5% in K-metric. For DBLP, consistent improvements were observed, with pP, pR, and pF1 increasing by around 9%, 7%, and 9%, respectively, with ACP, AAP, and K-metric by 2%.

These results indicate that using GCN in the framework with ComMAND improves the AND process with bibliographic datasets varying in scale, attributes, and level of ambiguity, reducing fragmentation and enhancing clustering quality.

5.4.2. Ablation 2

The SciBERT outperformed TF-IDF and Word2Vec across all datasets in this ablation variant, as shown in Table 5.

In AMiner-12, SciBERT demonstrated moderate improvements over TF-IDF, particularly in pP, pR, and pF1, with increases of around 7%, while outperforming Word2Vec significantly, with gains of up to 57% in pR and 39% in pF1. In AMiner-18, the larger and more text-rich dataset, the advantages of SciBERT were more pronounced, achieving improvements over TF-IDF of 45% in pP and over 70% in pF1 while also surpassing Word2Vec by up to 67% in pP. The differences in DBLP were less substantial, but SciBERT still showed improvements across all metrics, including pF1 with a 13% increase and around 4% in ACP, AAP, and K-metric compared to Word2Vec.

Considering the pF1 metric, SciBERT proved to be better than other types of embedding extraction, as shown in Fig. 2. These results highlight SciBERT's ability to leverage complex textual features effectively, particularly in larger datasets like AMiner-18 and DBLP. Its consistent performance across all metrics demonstrates its robustness and adaptability for AND tasks, making it a reliable approach for bibliographic datasets.

6. Results and analysis

In this section, we report the results of applying the framework with the datasets and evaluation metrics described in Section 5. A comparative analysis with baseline works is presented to illustrate the performance variations.

6.1. Baseline works

To validate the framework for AND, we compared consolidated works detected in the literature review. We selected baseline studies that employ similar techniques and use the same datasets for validation, ensuring an objective analysis of the advantages and limitations of our approach. According to these characteristics, the works of Pooja et al. (2020, 2021, 2022) and Zhou, Shi et al. (2024), described in Section 3 and presented in Table 1 were chosen as baseline. Additionally, we conducted a detailed verification to ensure that the comparison was fair and consistent, maintaining the same dataset configurations and evaluation metrics.

For a comparative analysis with the existing AND open-source framework, we include ANDez (Kim & Kim, 2024) as it follows a different approach from the other baselines. ANDez compares pairs of author mentions using similarity features based on metadata

Fig. 2. TF-IDF (green), Word2Vec (red), and SciBERT (blue) embedding methods on AMiner-12, AMiner-18, and DBLP datasets (x axis) with pF1 evaluation metric (y axis). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 6
General comparison between different baseline approaches on the AMiner-12 dataset.

Metric	Pooja et al. (2020)	Pooja et al. (2021)	Pooja et al. (2022)	Framework
pP	**0.836**	0.756	0.724	0.748
pR	0.578	**0.844**	0.751	0.738
pF1	0.621	**0.784**	0.715	0.733
AAP	0.909	0.856	0.798	**0.936**
ACP	0.647	0.782	0.810	**0.913**
K-metric	0.747	0.814	0.800	**0.924**

Table 7
General comparison between different baseline approaches on the AMiner-18 dataset.

Metric	Zhou, Shi et al. (2024)	Pooja et al. (2020)	Pooja et al. (2021)	Pooja et al. (2022)	Framework
pP	0.744	0.514	0.626	0.756	**0.778**
pR	0.753	0.599	**0.761**	0.671	0.587
pF1	**0.732**	0.554	0.669	0.694	0.636
AAP	–	0.694	0.799	0.768	**0.871**
ACP	–	0.626	0.674	0.729	**0.920**
K-metric	–	0.646	0.730	0.745	**0.895**

such as coauthors, title, venue, and author name. The framework applies traditional classifiers, such as random forest and logistic regression, to check whether two mentions refer to the same individual. After classification, the mentions are grouped into distinct author profiles using clustering algorithms such as agglomerative clustering or DBSCAN. ANDez offers modular components and supports reproducible experiments but does not incorporate semantic embeddings or graph-based learning. An initial comparison using ten ambiguous author names of the AMiner-12 dataset highlights the differences with our framework. The results of this comparison appear in Table 9 of Section 6.2.3.

6.2. Performance comparison

The results, along with the strengths and limitations of the proposed framework using ComMAND across different datasets, are presented in Tables 6, 7, and 8, with highlights on the best metric result. In the sequence, we analyze the results in each dataset, considering factors such as dataset size, attribute richness, graph density, and the number and size of ambiguous classes.

6.2.1. DBLP dataset

Considering the results of Table 8, the framework performs best on the DBLP dataset, consistently surpassing the baselines across all metrics. The richness and structure of the DBLP data, including detailed attributes such as titles, abstracts, publication venues, and co-authors, contribute significantly to this outcome. These attributes enable the construction of informative graphs and the extraction of meaningful embeddings. The GCN in the framework leverages relational and non-relational attributes effectively, delivering superior clustering quality metrics such as AAP, ACP, and K-metric.

The higher number of ambiguous classes in the DBLP dataset, compared to AMiner-12 and AMiner-18, also influences the results. While the increased number of classes adds complexity to the disambiguation task, the smaller size of each class in DBLP reduces

Table 8
General comparison between different baseline approaches on the DBLP dataset.

Metric	Pooja et al. (2020)	Pooja et al. (2021)	Framework
pP	0.789	0.853	**0.863**
pR	0.690	0.706	**0.880**
pF1	0.659	0.761	**0.869**
AAP	0.726	0.768	**0.970**
ACP	0.841	0.853	**0.973**
K-metric	0.754	0.807	**0.972**

Table 9
Performance comparison between the baseline works, the open-source ANDez (Kim & Kim, 2024), and this framework using ten ambiguous author names in AMiner-12.

Author	Pooja et al. (2020)			Pooja et al. (2022)			ANDez			Framework		
	pP	pR	pF1	pP	pR	pF1	pP	pR	pF1	pP	pR	pF1
Mark Davis	0.912	0.330	0.484	0.989	0.989	0.989	1.000	1.000	**1.000**	0.989	0.989	0.989
Koichi Furukawa	1.000	0.268	0.422	0.974	0.974	0.974	1.000	0.883	0.940	1.000	1.000	**1.000**
Fan Wang	0.991	0.796	0.883	0.881	0.937	**0.908**	0.379	0.750	0.533	0.862	0.874	0.868
Jose M. Garcia	1.000	0.578	0.732	0.863	0.974	0.915	0.702	0.542	0.617	1.000	1.000	**1.000**
Eric Martin	1.000	0.663	0.797	0.962	0.873	**0.915**	0.688	0.875	0.776	0.913	0.951	0.932
Jie Tang	1.000	0.806	0.892	0.905	0.966	0.935	0.523	1.000	0.723	0.967	0.999	**0.982**
Yue Zhao	0.917	0.771	0.837	0.876	0.968	0.920	0.383	0.889	0.584	0.914	0.978	**0.945**
Frank Mueller	1.000	0.231	0.376	0.980	0.980	0.980	1.000	1.000	**1.000**	0.979	0.940	0.959
David C. Wilson	1.000	0.865	0.928	0.999	0.999	**0.999**	0.859	0.859	0.859	0.998	0.966	0.981
Yoshio Tanaka	1.000	0.526	0.690	1.000	1.000	**1.000**	0.803	1.000	0.896	1.000	1.000	**1.000**

intra-class confusion, making it easier to distinguish authors. Combining high granularity (many small classes) and dense metadata enhances the effectiveness of our framework with ComMAND.

In contrast, baseline methods, such as Pooja et al. (2020), which rely on HAC and heuristics, struggle with larger numbers of ambiguous classes. These methods fail to fully exploit the richness of DBLP, which explains why our framework with ComMAND achieves relative improvements ranging from 1.1% to 33.6% across different metrics.

6.2.2. AMiner-18 dataset

The AMiner-18 dataset, characterized by its large scale and complexity, also showcases our framework strengths. AMiner-18 poses challenges related to ambiguous names and sparse connections. The framework's scalability and ability to incorporate textual attributes, such as keywords and abstracts, contribute to its strong performance, particularly in precision-related metrics, such as pP (0.778), and clustering quality metrics, like AAP (0.871), ACP (0.920), and K-Metric (0.895).

Although AMiner-18 has fewer ambiguous classes than DBLP, its sheer size and sparse relationships require methods capable of effectively handling large-scale data. The framework with ComMAND significantly improves clustering quality metrics. The ACP metrics show gains up to 78.0% over baselines. However, Zhou, Shi et al. (2024) on pF1 (0.732) and Pooja et al. (2021) on pR (0.761) outperform the framework, suggesting that their approach can better capture specific heuristics in the AND task. Nonetheless, our framework's superior performance in clustering metrics demonstrates its ability to handle the inherent complexity of large datasets.

6.2.3. AMiner-12 dataset

On the AMiner-12 dataset, which is smaller and less complex, our framework delivers performance comparable to the baselines. Simpler datasets , such as AMiner-12, benefit less from the advanced techniques used by ComMAND. For example, methods such as Pooja et al. (2020) achieve competitive results in metrics like pP (0.836). Pooja et al. (2021) also surpasses the framework in pR (0.844) and pF1 (0.784).

The lower number of ambiguous classes in AMiner-12 simplifies the AND task, reducing the advantage of sophisticated models like the framework. Nonetheless, our framework excels in clustering metrics, achieving the best results in AAP (0.936), ACP (0.913), and K-metric (0.924), with relative improvements ranging from 2.97% to 41.1%.

Table 9 presents the performance of our framework for ten ambiguous author names using the AMiner-12 dataset (Pooja et al., 2022). Our framework achieved the highest pF1 scores in 5 out of 10 cases, including Koichi Furukawa, Jose M. Garcia, Jie Tang, Yue Zhao, and Yoshio Tanaka. In names like Mark Davis and Frank Mueller, ANDez of Kim and Kim (2024) achieved perfect scores. Conversely, for Fan Wang, Eric Martin, and David C. Wilson, the proposal of Pooja et al. (2022) yielded better results. These variations reflect the distinct strengths of each approach, particularly the contrast between systems based on structural or metadata features and those integrating semantic and relational signals.

7. Discussion

In this section, we present a thorough discussion of the outcomes, including observations with the research's theoretical and practical implications. The results across all datasets demonstrate the adaptability of our framework to datasets with different sizes, attributes, and levels of ambiguity. Several key factors contribute to its superior performance:

Table 10
Execution time with a percentage per layer and total time for each dataset.

Dataset	Layer	Execution time (s/h)	Total time (%)
AMiner-12	L_1	8.37/0.002 h	0.4%
	L_2	366.90/0.102 h	17.8%
	L_3	1670.55/0.464 h	81.3%
	L_4	10.21/0.002 h	0.5%
	Total	**2056.03/0.571 h**	**100%**
AMiner-18	L_1	42.47/0.0118 h	0.5%
	L_2	2051.40/0.570 h	22.4%
	L_3	6968.58/1.935 h	76.0%
	L_4	105.68/0.029 h	1.1%
	Total	**9168.13/2.547 h**	**100%**
DBLP	L_1	15.95/0.004 h	0.8%
	L_2	436.15/0.121 h	21.7%
	L_3	1507.13/0.418 h	74.8%
	L_4	54.24/0.015 h	2.7%
	Total	**2013.47/0.559 h**	**100%**

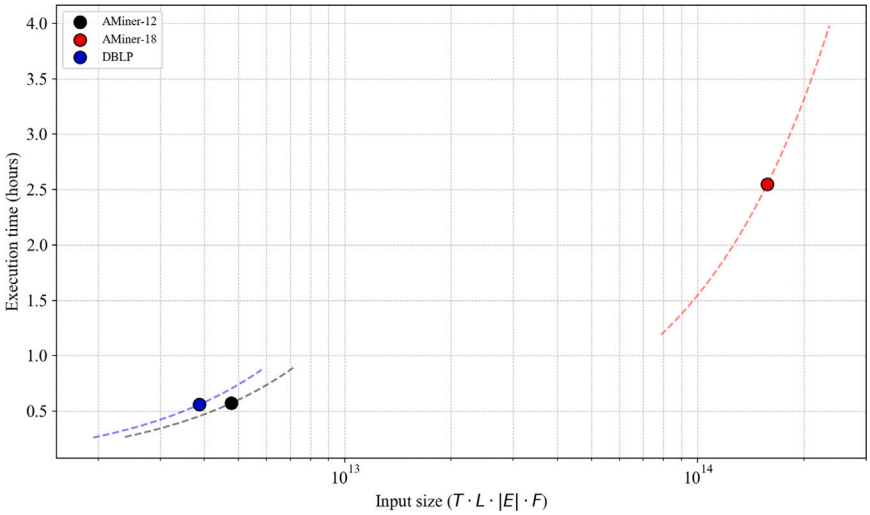

Fig. 3. Scalability illustration for L_3 with execution time (hours) vs. input size ($T \cdot L \cdot |E| \cdot F$) (logarithmic scale).

- Graph structure — in datasets with more structured relationships, such as DBLP, which includes keyword annotations and other attributes, the framework captures both semantic and topological information more effectively.
- Textual attributes — rich textual metadata, such as titles and abstracts in AMiner-18 and DBLP (see Table 2), enhances the embedding quality and clustering accuracy.
- Ambiguous classes — while DBLP presents more ambiguous classes than AMiner-12 and AMiner-18, its smaller class sizes reduce intra-class confusion, favoring the framework's precision in the AND task.

We validated the framework's scalability using the three datasets AMiner-12, AMiner-18, and DBLP, computed in the Google Colab L4 environment (GPU with 22.5 GB of RAM, CPU with 53 GB of RAM, and 201.2 GB of disk). Table 10 presents the execution times, in seconds (s) and hours (h), for each layer in the architecture shown in Fig. 1. The results indicate that L_3 consistently demands the largest share of the total execution time with the GCN training of heterogeneous graph, ranging from 74.8% to 81.3%. In contrast, the L_1 and L_4 layers require significantly less processing time, ranging from 0.9% to 3.5% of the total. This pattern remains consistent across the datasets, highlighting the increased complexity of the intermediate graph processing steps.

Fig. 3 with a logarithmic scale focuses on L_3, which is responsible for the bulk of computation in the framework. Results confirm that L_3 is the scalability bottleneck, with execution time increasing consistently with graph complexity, as expected from the theoretical cost $\mathcal{O}(T \cdot L \cdot |E| \cdot F)$ discussed in Section 4.1.3. AMiner-18 required the most time (1.935 h), reflecting its larger graph structure. AMiner-12 executed faster (0.464 h). DBLP took (0.418 h), even with more detailed input data. The comparison shows that graph size and connectivity, rather than the richness of node attributes, are the primary factors impacting performance. As expected, AMiner-18 had the longest total execution time (2.547 h), AMiner-12 (0.571 h), and DBLP (0.559 h), which is consistent with their graph sizes. The framework's scalability indicates opportunities for further efficiency improvements.

Table 11 compares our framework to the best scores reported by baseline methods (boldface). Our framework performs strongly across classification and clustering metrics, surpassing baselines in nearly all clustering results (AAP, ACP, K-metric). The DBLP

Table 11
Comparison of the proposed framework with baselines across datasets.

Dataset	Metric	Task	Framework	Baselines
AMiner-12	pP	Classification	0.748	**0.836** (Pooja et al., 2020)
	pR	Classification	0.738	**0.844** (Pooja et al., 2021)
	pF1	Classification	0.733	**0.784** (Pooja et al., 2021)
	AAP	Clustering	**0.936**	0.909 (Pooja et al., 2020)
	ACP	Clustering	**0.913**	0.810 (Pooja et al., 2022)
	K-metric	Clustering	**0.924**	0.814 (Pooja et al., 2021)
AMiner-18	pP	Classification	**0.778**	0.756 (Pooja et al., 2022)
	pR	Classification	0.587	**0.761** (Pooja et al., 2021)
	pF1	Classification	0.636	**0.732** (Zhou, Shi et al., 2024)
	AAP	Clustering	**0.871**	0.799 (Pooja et al., 2021)
	ACP	Clustering	**0.920**	0.729 (Pooja et al., 2022)
	K-metric	Clustering	**0.895**	0.745 (Pooja et al., 2022)
DBLP	pP	Classification	**0.863**	0.853 (Pooja et al., 2021)
	pR	Classification	**0.880**	0.706 (Pooja et al., 2021)
	pF1	Classification	**0.869**	0.761 (Pooja et al., 2021)
	AAP	Clustering	**0.970**	0.768 (Pooja et al., 2021)
	ACP	Clustering	**0.973**	0.853 (Pooja et al., 2021)
	K-metric	Clustering	**0.972**	0.807 (Pooja et al., 2021)

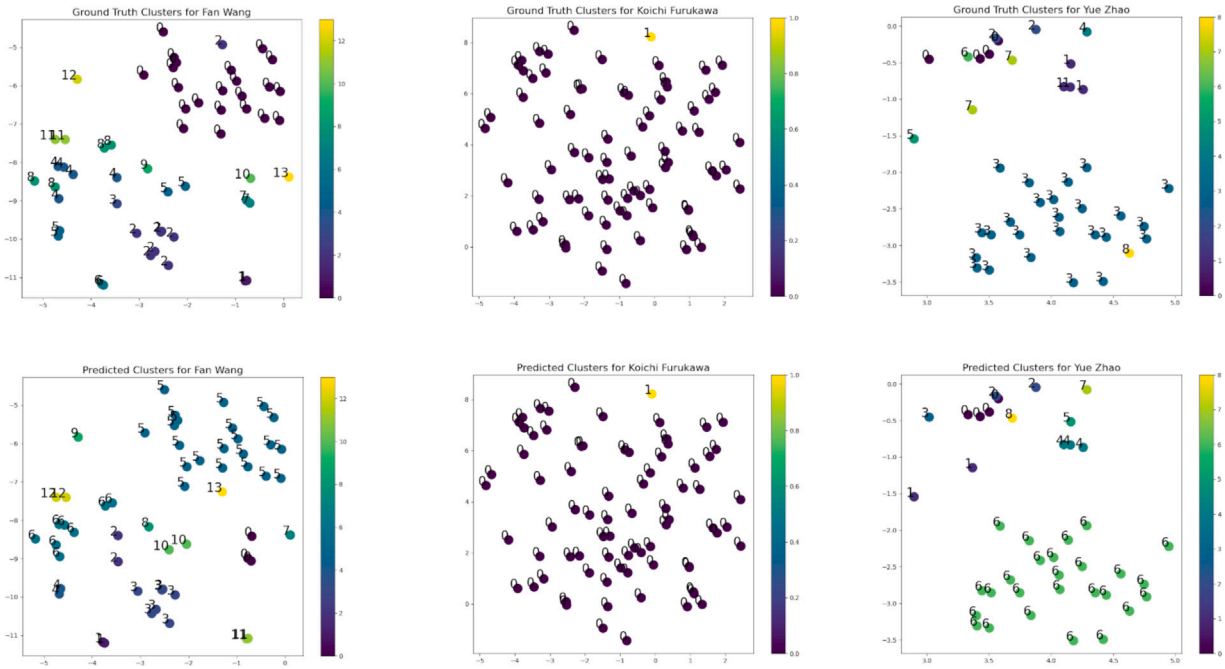

Fig. 4. Ground truth vs. predicted clusters from three distinct authors from the AMiner-12 dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

dataset achieves the highest values across all metrics, benefiting from dense relationships, rich metadata, and many small ambiguous classes. In AMiner-18, it outperforms baselines in clustering and achieves the best pP. Although baselines outperform our method on pR and pF1 in AMiner-12, our framework still leads in clustering. These results confirm its effectiveness in identifying document groupings and structural patterns essential to the AND task.

Fig. 4 presents clustering results for three ambiguous authors from the AMiner-12 dataset: Fan Wang (numbered from 0 to 13), Koichi Furukawa (numbered from 0 to 1), and Yue Zhao (numbered from 0 to 8). These authors have 14, 2, and 9 ground truth labels, respectively, with corresponding pF1 values of 0.868, 1.00, and 0.945 (as reported in Table 9 for our framework). The disambiguation task becomes increasingly challenging as the number of associated clusters grows. For example, Fan Wang, who has the highest number of ambiguous instances, exhibits more fragmented predictions and a lower pF1 score compared to the other two authors.

In addition, Fig. 4 illustrates the visual outcome of the clustering process for these authors. Each column represents an ambiguous name, with the top and bottom rows showing the ground truth and the predicted clusters, respectively. Publications are grouped and

color-coded according to their cluster assignments, which allows for a clear visual comparison. Even though the cluster indices may differ from the correct labels, the overall grouping structure is often consistent. This visualization demonstrates the model's capacity to recover distinct author identities and reveals how ambiguity influences performance. Authors associated with fewer classes, such as Koichi Furukawa, typically result in compact and coherent clusters, while those with a more significant number of classes, like Fan Wang, present more fragmented outputs.

In contrast, the variability in results across datasets highlights some of our framework's limitations:

- Simpler datasets — in smaller and less complex datasets like AMiner-12, baselines that rely on heuristics or HAC achieve competitive results. The fewer ambiguous classes and less dense graphs limit the advantages of advanced models like our framework.
- Dependence on textual features — the reliance on textual features, such as abstracts and keywords, may limit our framework effectiveness in datasets with sparse or incomplete attributes. In these cases, relational features alone may not provide sufficient discriminative power.
- Class distribution — while smaller class sizes in DBLP facilitate the AND task, highly imbalanced class distributions (e.g., many small classes versus a few large ones) in other datasets could pose challenges. Such challenges may require additional balancing mechanisms or pre-processing steps.

The results suggest that the framework with ComMAND is particularly well-suited for datasets like DBLP, where the combination of a high number of small classes, dense graphs, and complex textual metadata creates an ideal environment for its GCN-based approach. In larger datasets like AMiner-18, our framework demonstrates scalability, excelling in precision and clustering metrics despite the sparse and complex nature of the data. The adaptability to datasets with varying characteristics further highlights our framework's robustness and effectiveness in addressing the AND problem.

Recent advances in using LLMs, as demonstrated in Yan and AsirAsir (2024), offer promising avenues for improving AND tasks. Integrating LLMs to capture more complex semantic and contextual relationships could enhance our framework, especially on challenging datasets with sparse connections and ambiguous relations. However, further experiments are necessary to evaluate the effectiveness of such use.

7.1. Theoretical and practical implications

The proposed novel framework for AND brings a different theoretical contribution. Multi-model machine learning techniques for AND applications resemble ensemble learning. Our approach focuses on meaningful contextual semantic learning using transfer learning with SciBERT, combining structural and textual features with GCN and GHAC. Existing research applying graph networks alone is limited to specific patterns of authors and publications. While GCN can uncover hidden relationships, it can also model co-authorship, citation links, and other network features vital for accurately identifying authors. The association of GCN with GHAC permits heterogeneous hierarchical clustering with graph methods to boost author identification. Thus, the limitations of graph networks research, language model identification learning, and graph clustering applied to the AND are lightweight by the proposed ComMAND method.

8. Conclusion

This work presented a novel framework with ComMAND for addressing the AND challenge in digital bibliographic repositories by combining semantic content analysis, graph-based modeling, and clustering techniques.

Experimental results on datasets such as AMiner-12, AMiner-18, and DBLP highlight the superior performance of our framework. On DBLP (Table 8), it achieved a pF1 of 0.869 and a K-metric of 0.972, with improvements from 1.1% to 33.6% over baseline works (Pooja et al., 2020, 2021), showcasing its effectiveness in large-scale AND tasks.

The framework with ComMAND demonstrates strong performance across diverse datasets, positioning itself as a robust tool for advancing person-centric bibliographic repositories. While promising, our approach faces limitations in less-structured repositories that need more rich metadata. Future work could explore performance comparison using larger datasets, such as LAGOS-AND (Zhang et al., 2023). The investigation of new techniques using graph imputation, multilingual embeddings, and the integration of LLMs to improve the framework's adaptability. Techniques such as distributed graph processing, graph sampling, embedding reuse, dimensionality reduction, and incremental GCN training for real-time updates could enhance scalability.

CRediT authorship contribution statement

Natan S. Rodrigues: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Célia G. Ralha:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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